



HEMATOLOGY AND BIOCHEMICAL INTERPRETATIONS- BASICS

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Learning is a Continuous Process





SAMPLES NEEDED

- WHOLE BLOOD
- SERUM
- BUFFY COAT SMEAR
- URINE

- 
- **BLUE** –SODIUM CITRATE
 - **RED**- COT ACTIVATOR / FREE
 - **PURPLE /LAVENDER** – EDTA
 - **GREEN** – HEPARIN
 - **GREY** – SOD FLUORIDE

VACUTAINERS

Tube cap color	Additive	Function of Additive	Common laboratory tests
Light-blue 	3.2% Sodium citrate	Prevents blood from clotting by binding calcium	Coagulation
Red or gold (mottled or "tiger" top used with some tubes is not shown) 	Serum tube with or without clot activator or gel	Clot activator promotes blood clotting with glass or silica particles. Gel separates serum from cells.	Chemistry, serology, immunology
Green 	Sodium or lithium heparin with or without gel	Prevents clotting by inhibiting thrombin and thromboplastin	Stat and routine chemistry
Lavender or pink 	Potassium EDTA	Prevents clotting by binding calcium	Hematology and blood bank
Gray 	Sodium fluoride, and sodium or potassium oxalate	Fluoride inhibits glycolysis, and oxalate prevents clotting by precipitating calcium.	Glucose (especially when testing will be delayed), blood alcohol, lactic acid

	UNITS	CANINE	FELINE	BOVINE	EQUINE
ERYTHROCYTES	(× 10 ⁶ /μl)	5-8.5	5-10	5-10	6.5-12.5
HAEMOGLOBIN	(g/dl)	12-18	9-16	8-15	11-19
HEMATOCRIT	(%)	35-55	28-45	24-46	32-52
MCV (MEAN CORPUSCULAR VOLUME)	(fl)	60-77	39-55	40-60	34-58
MCH (MEAN CORPUSCULAR HAEMOGLOBIN)	(pg)	21-27	13-17	11-18	10-18
MCHC (MEAN CORPUSCULAR HAEMOGLOBIN)	(g/dl)	32-36	30-36	30-36	30-35

	UNITS	CANINE	FELINE	BOVINE	EQUINE
PLATELETS	($\times 10^3/\mu\text{l}$)	200-700	200-700	100-800	100-600
TOTAL LEUKOCYTES	($\times 10^3/\mu\text{l}$)	6-17	5.5-19.5	4-12	5.5-12.5
NEUTROPHILS	($\times 10^3/\mu\text{l}$)	3-11.5	2.5-12.5	0.6-4	2.7-6.7
LYMPHOCYTES	($\times 10^3/\mu\text{l}$)	1-4.8	1.5-7	2.5-7.5	1.5-3.5
MONOCYTES	($\times 10^3/\mu\text{l}$)	0.1-1.4	0-0.8	0-0.8	0-0.8

	UNITS	CANINE	FELINE	BOVINE	EQUINE
EOSINOPHILS	($\times 10^3/\mu\text{l}$)	0.1-1.2	0-1.5	0-2.4	0-0.9
BASOPHILS	($\times 10^3/\mu\text{l}$)	Rare	Rare	0-0.2	0-0.2
ERYTHROCYTE SEDIMENTATION RATE	(minutes)	5.0-25	—	—	51-63



INTERPRETATION OF HEMATOLOGICAL VALUES

HAEMOGLOBIN

- **Increased**- Polycythemia , dehydration, high altitude.
- **Decreased** - Anemia

HEMATOCRIT

- **Increased** - Haemoconcentration, polycythemia
- **Decreased** - Bone marrow depression, Anaemia

PLATELETS

- **Increased** -Thrombocytosis- associated with inflammation, myeloproliferative disease.
- **Decreased** – Thrombocytopenia, IMHA
IMT, E.canis



TOTAL LEUKOCYTES

- **Increased** - Leukocytosis- leukemia, inflammatory disease, exercise, fear, near term of pregnancy, day of estrus.
- **Decreased**- Viral diseases due to
- neutropenia



LYMPHOCYTES

- **Increased** - Infectious hepatitis, bacillary infections, tuberculosis, goiter, lymphoma, lymphocytic leukaemia.
- **Decreased** - Viral infection, immunosuppressive drugs, radiation.

MONOCYTES

- **Increased** -Chronic infection, TB, subacute bacterial endocarditis, brucellosis, swine erysipelas, listeriosis, **E.canis**, monocytic and myelomonocytic leukemia, relative increase in leukopenia, neutropenia.
- **Decreased** – Not seen

NEUTROPHILS

- **Increased** - Acute infections, pneumonia, metritis, pyometra, meningitis, rheumatoid arthritis, localized pyaemic infections, like abscesses, furuncle, otitis, intoxication like uraemia, gout, diabetic acidosis, drug poisoning like lead, mercury, insect venoms, septicaemia, toxaemia,



NEUTROPHILS

- **Decreased** - Bacterial infection, endotoxaemia, immune mediated diseases, cancer chemotherapeutic drugs, radiation, idiosyncratic drug reactions, neoplasia.

EOSINOPHILS

- Allergic conditions, skin diseases, parasitic diseases, eosinophilic myositis, radiation therapy, tumours of ovary, serous surfaces and bones, eosinophilic leukaemia, poisons like copper sulphate, phosphorus, camphor.
- Epinephrine treatment, corticosteroids, acute infection.

BASOPHILS

- **Increased** - Chronic sinus inflammation, chronic haemolytic anaemia, foreign protein infection, Hodgkin's disease, basophilic leukaemia, granulocytic leukaemia.
- Not seen



NORMAL SERUM BIOCHEMICAL VALUES

	UNITS	CANINE	FELINE	BOVINE	EQUINE
Glucose	(mg/dl)	75-115	85-115	60-100	70-100
Cholesterol	(mg/dl)	130-240	90-160	—	—
Total protein	(g/dl)	5.7-7.4	6.0-8.1	6.4-8.2	6.0-8.0
Albumin	(g/dl)	2.9-4.0	2.9-3.5	3.1-4.0	3.0-4.1
Globulin	(g/dl)	2.1-4.1	2.8-4.9	2.7-4.5	2.0-4.4
BUN (BLOOD UREA NITROGEN)	(mg/dl)	6-22	15-30	10-24	14-27
Creatinine	(mg/dl)	0.4-1.5	0.6-2.2	0.7-1.8	1.2-2.5

	UNITS	CANINE	FELINE	BOVINE	EQUINE
Bilirubin	(mg/dl)	<0.4	<0.2	<0.5	<2.0
ALT (Alanine amino transferase)	(IU/L)	<100	<90	—	—
AST (Aspartate amino transferase)	(IU/L)	<60	<40	<150	<350
ALP (Alkaline phosphatas e)	(IU/L)	<135	<45	<140	<300
GGT (Gam maglutamyl transferase)	(IU/L)	—	—	<45	<45

	UNITS	CANINE	FELINE	BOVINE	EQUINE
Amylase	(IU/L)	<900	<900	—	<20
Lipase	(IU/L)	<600	<400	<50	<20
Anionic gap	(mmol/ L)	8-20	8-20	6-15	6-15
Total co₂	(mmol/L)	20-28	20-28	21-31	24-30
calcium	(mg/dl)	8.6-11.2	8.9-11.0	8.5-11.1	10-13
phosphorus	(mg/dl)	2.5-5.5	3.1-5.8	4.0-7.2	1.9-4.7
Sodium	(mmol/L)	140-155	140-155	135-150	130-145

	UNITS	CANINE	FELINE	BOVINE	EQUINE
Potassium	(mmol/L)	3.5-5.5	3.0-5.5	3.5-5.0	3.5-5.0
Magnesium	(mmol/L)	—	—	0.70-1.10	0.60-0.95
Chloride	(mmol/L)	105-120	110-125	95-110	95-110

TOTAL PROTEIN

- Normal : 7 gm/dl
- Albumin : 4gm/dl
- Globulin : 3gm/dl

Total protein



- Overhydration,
- Severe congestive heart failure (with edema),
- Protein-losing nephropathy,
- Protein-losing enteropathy,
- Hemorrhage, burns,
- Dietary protein deficiency,
- Malabsorption

Total Protein



- Dehydration,
- Chronic inflammation
- Paraproteinemia
- Increase intake of Protein



Renal function

- BUN : 10-30 mg/dl
- Creatinine : 0.5 – 1.5 mg/dl
- Urine protein : Creatinine ratio
UP/C

IRIS* Classification of Canine Chronic Kidney Disease (CKD)

STAGE mg/dl	I Non Azotemia	II Mild	III Moderate	IV Severe Azotemia
Creatinine	<1.4	1.4-2.0	2.1-5.0	>5
ne				

Urine protein : Creatinine ratio

Urine Protein / Creatinine ratio	Stage
<0.2	Non-proteinuric
0.2 to 0.5	Borderline proteinuric (BP)
>0.5	Proteinuric (P)



Liver Function tests

- Alanine aminotransferase (ALT)
- Aspartate aminotransferase (AST)
- Serum alkaline phosphatase (SAP)
- Gamma-glutamyltransferase (GGT)
- Total Protein

Pancreas

- Amylase,
- Lipase,
- Glucose
- TLI



Electrolytes

- Sodium
- chloride,
- potassium,
- phosphorus,
- calcium,
- magnesium

Phosphorus

- Increased - chronic kidney disease.
Dogs when about 75 percent of
both kidneys is damaged.



Potassium

- **Increased** - acute kidney failure, ruptured or obstructed bladder.
- **Decreased** - Low blood potassium can cause the pet to feel weak.
Cats with low potassium may develop painful muscles.
Addison's disease



Sodium

- **Increased** - dehydration

Decreased - Addison's disease
(hypoadrenocorticism).

Glucose

- Normal – 80-116 mg/dl
cat – 60-150mg/dl
- **Increased** in dogs and cats - diabetes mellitus..

Decreased - pancreatic cancer, sepsis .
Low blood sugar can cause depression or seizures..



Hb I Ac (Glycated Hemoglobin)

- Normal- 1.7 % to 4.9%
- Whole blood in EDTA tube

Bile acids

- **Bile acids** are produced by the liver and are involved in fat breakdown. A bile acid test is used to evaluate the function of the liver and the blood flow to the liver. Patients with abnormal blood flow to the liver, a condition known as **portosystemic shunt** will have abnormal levels of **bile acids**. The bile acid test measures a fasting blood sample and a blood sample two hours after eating.



Bilirubin

- Normal upto 0.2 mg/dl
- Produced by the liver from old red blood cells. Bilirubin is further broken down and eliminated in both the urine and stool.
- Icterus,
- intravascular hemolysis,
- Biliary obstruction

ALT

- Normal 16-73 iu/L
- Assessment of liver disease
- Not a liver function test
- **Increased** - hepatocyte injury, Acute hepatitis, neoplasia, cirrhosis,
Decreased – end stage liver disease

SAP

- Normal 15-146 iu/L
- **Increased** – obstructive liver/biliary tract
not specific for LIVER
young ones (growing bone)
bone diseases, osteosarcoma
hyperphosphatemia,
acute pancreatitis, hepatitis

Corticosteroid therapy

Amylase

- Normal 347-1104 iu/L
- is an enzyme produced by the pancreas and the intestinal tract.
- Amylase helps the body breakdown sugars.
- Increased (pancreatitis) or cancer of the pancreas..
- Amylase clearance depends on Renal function. CKD



Lipase

- Normal 22-216 iu/L
- Lipase is another pancreatic enzyme which is responsible for the breakdown of fats and which may be increased in patients with pancreatic inflammation or cancer.

Anion gap

- $\text{Na} - (\text{Cl} + \text{HCO}_3)$
- Normal 16.3 -28.6



General Metabolism

GLU (Glucose)
LDH (Lactate
dehydrogenase)
CPK (Creatine
phosphokinase)

Kidney Function

BUN (Blood Urea
Nitrogen)
CREAT (Creatinine)

Electrolytes

Na (Sodium)
K (Potassium)
Cl (Chloride)
CA (Calcium)
PHOS (Phosphorus)



Liver Function	Thyroid	Pancreas
ALP (Alkaline phosphatase)		
ALB (Albumin)		
GGT (Gamma-glutamyl transpeptidase)		
SGPT (Serum glutamate pyruvate transaminase)	T3 (Triiodothyronine)	AMY (Amylase)
TP (Total Protein)	T4 (Thyroxine)	LIP (Lipase)
CHOL (Cholesterol)		
GLOB (Globulin)		
TBILI (Total Bilirubin)		

COW

- **GLUCOSE** - 45-70 mg/dl
- **CALCIUM** – 9-12 mg/dl
- **PHOSPHORUS** – 5-6 mg/dl
- **POTTASIUM** - 4-5 meq /dl
- **MAGNESIUM** -1-2 mg/dl
- **Cpk** – 35-280 u/L

THANK YOU

